

SOME ARCHAEOLOGICAL OBSERVATIONS ON THE AQUEDUCTS OF QUMRAN

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De Vaux dated the building of the main aqueduct to before the earthquake of 31 BCE,¹ a dating subsequently accepted by others,² although it has been assigned, without discussion, to the time of Herod Archelaus,³ and, with little explanation, it has been claimed that “the entire water supply system was constructed at one time, that is, in the Herodian period.”⁴ A careful review of de Vaux’s field notes and of the published photographs⁵ together with observations made in the field⁶ reveal that the aqueduct can be dated no earlier than the time of Herod after the earthquake. To justify this dating, which is of importance for an understanding of Qumran, one must consider the known evidence for the earliest, post-Iron Age building phase (de Vaux’s “Period Ia”) that pre-dated the “Main” aqueduct, and how it was altered by the construction of that aqueduct. De Vaux identified earlier (pre-main aqueduct) channels which fed pools loci 117, 110 and 118 via a sed-

¹ Roland de Vaux, *Archaeology and the Dead Sea Scrolls* (Oxford: Oxford University Press, 1973) 7–10.

² E.g., Jodi Magness, *The Archaeology of Qumran and the Dead Sea Scrolls* (Grand Rapids, MI: Eerdmans, 2002) 67–68.

³ Z. Ilan and D. Amit, “The Aqueduct of Qumran,” in *The Aqueducts of Israel* (eds. D. Amit, J. Patrich, and Y. Hirschfeld; Journal of Roman Archaeology Supplement 46; Portsmouth, RI: Journal of Roman Archaeology, 2002) 385.

⁴ Y. Hirschfeld *Qumran in Context* (Peabody, MA: Hendrikson, 2004) 115.

⁵ All plates referred to, unless otherwise attributed, are from J.-B. Humbert and A. Chambon, *Fouilles de Khirbet Qumran et de 'Ain Feshka Vol I* (Göttingen: Academic Press Fribourg, 1994); references to loci with the date of the original note are from the English translation of S. Pfann, *The Excavations of Khirbet Qumran and Ain Feshka: Synthesis of Roland de Vaux's Field Notes Vol Ib*. (Göttingen: Academic Press Fribourg, 2003)

⁶ David Amit and Rachel Bar-Nathan of the IAA facilitated my visit on 27/5/05. We were accompanied by Ian Hutchesson who has generously supplied photographs for this article. I would like to thank them all for sharing their considerable knowledge of Qumran and making the visit both illuminating and pleasurable. They do not, necessarily, agree with all of my interpretations.

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imentation pool L 119⁷ but they are poorly recorded and scarcely appear in the photographic records.

A review of the earliest phase(s) in loci near the cistern L 110 (from south to north) Cf. Fig. 1

In L 100 de Vaux noted “the lower floor continues under the channel. The same floor reappears in the western part of L 106, where it is clearly cut by the main channel. . . . [. . .] This floor seems to be connected with the northern wall of L 106, and with the west and south walls of L 100” (L 100, notes 20/3/55). This floor (clearly that shown with a ranging rod lying on it in Pl 293) relates to the lowest point of plastering on the “bench” in the alcove in the southern wall. In L 106 it was noted that the north wall of L 106 and L 101 was “cut by the channel” (L 106, 14/2/55). Thus all of this “space” (loci 100 & 106) pre-dates the “main” aqueduct. Because no clear eastern wall was noted, it was probably an open courtyard. From it one could enter loci 104, 102 and 101 (“a lower floor of beaten earth, analogous to the lower floors of L 106 and 109, and which was earlier than the system of channels” L 101, 24/3/55). In the northwest corner of the courtyard, in the angle between the east wall of L 101 and the north wall of loci 101/106 the early, pre-aqueduct channel was detected below the main aqueduct starting immediately north of the north wall of loci 100/106 and directing water into L 117 (L 119 bis, 3/4/55). Water was collected on the surface of the courtyard and/or the roofs above L 101 and loci 110 + L 117.

Loci 109/113 could be entered from L 101. In L 109 “over the entire locus we uncovered a lower floor, earlier than the main channel and undoubtedly contemporary with the first circular cistern” (L 109, 23/3/55); and, in L 113 “We uncovered the original plaster floor which connects by a narrow opening with the channel feeding⁸ cistern 110” (L 113, 22/3/55) and “exposed the widest lip from the first stage of cistern 110” (L 113, 24/3/55). The staircase in L 113 was clearly built to a later floor associated with the building of the aqueduct and the raising of the sides of L 110 (Pl 233).

L 114. Only one floor was noted here. A plastered floor (L 114, 10/2/55) laid almost on virgin soil (L 114, 28/3/55). In the fill or accumulation

⁷ De Vaux, *Archaeology* 4; Humbert and Chambon *Fouilles de Kh Qumran*, Plan III; oddly this plan was omitted by Pfann.

⁸ Later identified as an overflow from, rather than an inlet to, L 110.

above this floor were many pottery vessels. These were photographed from the west, (Pls 222, 223). In Pl 222 the stones at the top of the photo must be those of the main channel and those at the bottom those of the cistern, L 110. Pl 223 appears to cover the area immediately to the south of Pl 222 and, again, the stones at the top must be from the aqueduct. The pottery runs beneath, and thus pre-dates, the aqueduct. The plaster floor related to the first stage of L 110 (as implied in 10/2/55) before its sides were raised (cf Pls 225, 226).

L 115. A “solid plastered floor with a ridge” (L 115, 9/2/55) can be seen in Pls 211, 212 & 215 which pre-dated the main aqueduct. Beneath it ran the pre-aqueduct channel. In the pre-aqueduct stage the space, loci 115/116, was a large room because the wall that divided it in two was not bonded with the western wall (Pl 211) and was probably built contemporaneously with the main aqueduct (Pl 215). The room could only have been entered from L 114 through a door at the eastern end of the separating wall (indicated on plan XVII).

L 116. “We found the lower channel in the northwest corner” (24/3/55). The entrance to this channel was in L 130 (Pl 269) where, apparently, “under the wall, there was a basin” (L 130, 24/3/55). Although de Vaux says that it “collect(ed) the water which drained on to the esplanade to the north of the ruins,”⁹ he did not define precisely where and on what surface, nor is it possible to detect how water was steered into the channel before the block of rooms, loci 120, 111 etc. were added (abutting walls of loci 120 and 116 visible in Pl 269). It is possible that most of the area later taken up by loci 130–2 and 135 was originally a large rain-catchment area, much of which was plastered [plastered surface in east of L 130 (L 130, 3/3/55); plaster in south of L 132 (L 132, 7/3/55)], draining into the lower channel.

The three loci 114, 115 and 116 are crucial for dating and understanding the main aqueduct. It would appear that de Vaux considered that the “Main” aqueduct, as it “wound its way between the buildings,”¹⁰ was originally free standing. Even though he was aware that “this channel was at a higher level than that of Period Ia” and that that necessitated the raising of the walls of the round cistern, L 110,¹¹ Humbert and Chambon certainly show the aqueduct co-existing with the original (“Period Ia”) floors in loci 114, 115, 116 and 100

⁹ De Vaux, *Archaeology*, 4.

¹⁰ De Vaux, *Archaeology*, 8.

¹¹ De Vaux, *Archaeology*, 9. At the same time the sides of the pools, L 117 and 118, would have had to have been raised, and three or four steps added.

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(Plans XVII and XXII, Period Ib); this despite the fact that, as noted above, in L 100 “the lower floor continues under the channel” and that the same floor in the western part of L 106 “is clearly cut by the main channel,” i.e., the floor did not co-exist with the aqueduct. Moreover the aqueduct ran through the full width of the doorway between loci 114 and 115. The only access to L 115 was through this door and, once the channel was built, could only have been via a floor covering the channel itself. As there are no signs that there were ever steps going up the (at least) 75 cm (see metre stick in Pl 211) from the plastered floors in loci 114 or 115 to a floor over the cap-stones covering the top of the main aqueduct, we must accept that the plastered floor pre-dated the main aqueduct and that any floor in these two rooms associated with the aqueduct was flush with the stone slabs that would have covered it, and with the top of the raised sides of L 110. According to de Vaux’s interpretation most of the raised south and west sides of the pool would have been free standing until his Period II. A substantial pool holding water at a higher elevation than the external floor surface would, to my knowledge, be unique in this period. Certainly none were found in Jericho, Cypros, Herodium or Masada.¹² When the sides were raised the area between the pool and the walls to its west and south would have been backfilled to help withstand the weight of the water in the pool. Thus the ceramics in L 114 pre-date the construction of the main aqueduct.

If the main aqueduct had been free-standing it would have considerably hampered movement around the site. In Jericho many aqueduct systems from a similar period have been thoroughly excavated. On open ground they were built as free standing channels between two well built and well plastered side walls. All three faces of the side walls were well plastered.¹³ Within occupied areas side channels led off the main channel to take water to particular venues, but these feeder channels were generally built conveniently into the ground beneath floors; the side walls of the channels were built with rough retaining walls which were plastered only on their inner faces and partially over the top. The channels were then covered with stone slabs.¹⁴

¹² Built bathtubs holding enough water for individual bathing were found at Jericho and Masada but even they had one long side attached to a wall.

¹³ See, *inter alia*, E. Netzer, *Hasmonean and Herodian Palaces at Jericho Vol I* (Jerusalem: IES, 2001) Ill. 111; E. Netzer *Hasmonean and Herodian Palaces at Jericho Vol II* (Jerusalem: IES, 2004) Ills 23–25.

¹⁴ See in particular in the Industrial zone Netzer, *Palaces Vol II*, Ills 56–62.

The same system would have been employed in Qumran. In a photograph of the aqueduct crossing L100 taken in 1956 soon after the excavations,¹⁵ and in other more recent pictures,¹⁶ there is no evidence that the aqueduct was ever free standing. The side walls are of irregular thickness, are haphazardly built, and have no clear outer face. Observation in the field convince that these walls could not have been freestanding but that they represent the facing walls for a sunken aqueduct associated with a floor at the elevation of the top of these walls (i.e. the floor in L 100 associated with the mill, Pls 295, 300, defined as belonging to Period II).

When was the main aqueduct constructed?

We have seen that the main aqueduct was built at a higher elevation than the earlier water system which had been buried by debris 75 cms or more deep. In some of the other pre-aqueduct rooms the debris appears to have been even deeper: e.g., loci 122 and 123 appear to be sub-floor storage bins dug at least a metre into the upper floor of L 121 (Pl 201). Such a depth is unlikely to represent ordinary occupation debris but could, at least in part, have been caused by the earthquake known to have occurred in 31 BCE. It was easier to build the aqueduct into this debris than to clear it, particularly as an integral part of the aqueduct system was the digging of the large pool, L 91. This would have generated a lot of spoil that had to be “lost” nearby and would have been ideal for levelling and creating new floors.

Unfortunately no topographic plan of Qumran has been published. Some rough elevations shown on a plan of a limited area¹⁷ does suggest that the ground level at the southern end of the site was some 40 cm higher than that to the north. Thus it may be that the main aqueduct either “needed to be elevated” or “was now able” to cross a shallow depression in order to reach the slightly higher ground into which L 91 was dug.

We have shown that the pottery from L 114 must pre-date the construction of the main aqueduct. To give a precise dating from ceramic evidence is difficult; in the case of Qumran it is even harder because it “has served as a linchpin for Palestinian ceramic typology and

¹⁵ [Http://www1.ku-eichstaett.de/KTF/qumran/bilder/abfrage/loc100.jpg](http://www1.ku-eichstaett.de/KTF/qumran/bilder/abfrage/loc100.jpg)

¹⁶ Hirschfeld, *Qumran*, figures 44, 58.

¹⁷ De Vaux, “Fouilles au Khirbet Qumran: Rapport préliminaire sur la deuxième campagne,” *RB* 61 (1954) Pl V.

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chronology of the Hasmonean and Herodian periods. Many of the types from Qumran are illustrated in Paul Lapp's landmark handbook.¹⁸ Lapp was no doubt influenced by his knowledge of the Qumran pottery when he published other key sites such as Beth-Zur¹⁹ and Tel el-Ful.²⁰ Thus a certain amount of circularity has crept into the dating of pottery from these periods. However a date of c. 31 BCE is not unreasonable for all of the pottery published from loci 114, although some doubt might be caused by the presence of three so-called "Herodian" Lamps²¹ which might be considered a little later. Bar-Nathan, who has published the pottery from Jericho, regards 37 BCE as too early for these lamps (her J-LP4) but she admits that they "had appeared by the Herodian 2 period (15 BCE–6 CE) alongside mold-made lamps, especially those with the spatulate nozzle, J-LP2C."²² As she also conjectures that a rare intermediate wheel made lamp, J-LP3, found only in Jericho, Qumran²³ and Masada, can be "associated with a pottery workshop in the Jericho valley and Dead Sea area,"²⁴ and, as we know that there were pottery kilns in Qumran, it is possible that the "Herodian" lamp was pioneered in Qumran, so a date for its appearance there just before the earthquake of 31 BCE seems justifiable.

De Vaux's mis-dating of the aqueduct, accepted with little question by Magness, has led the latter to further confuse the issue. Looking for loci whose pottery she believes can be securely dated to the earthquake of 31 BCE she chooses loci 49, 50 and 89, none of which were built until after the aqueduct—and, in the case of L 89, not until at least two sub-phases later (see below). She also mistakenly dates the pottery from L 130 to 9/8 BCE²⁵ and by comparing vessels from L 89

¹⁸ J. Magness, *Debating Qumran, Collected Essays on its Archaeology* (Leuven: Peeters, 2004) 1. See P.W. Lapp, *Palestinian Ceramic Chronology, circa 200 B.C.–A.D. 70* (New Haven, CT: ASOR, 1961).

¹⁹ P.W. Lapp, "Bethel Pottery of the Late Hellenistic and Early Roman Periods," *The Excavation of Bethel (1934–1960)* (ed. James L. Kelso; AASOR 39; Cambridge, MA: ASOR, 1968), 77–80.

²⁰ P.W. Lapp, *The Third Campaign at Tell el-Ful: The Excavation of 1964*. (AASOR 45; Cambridge, MA: ASOR, 1978).

²¹ R. de Vaux, "Fouilles au Khirbet Qumran: Rapport préliminaire sur les 3e, 4e et 5e campagnes," *RB* 63 (1956) Fig 4:14.

²² Rachel Bar Nathan, *Hasmonean and Herodian Palaces at Jericho: Vol III, The Pottery* (Jerusalem: IES, 2002) 112–113.

²³ De Vaux, *RB* 63, Fig 1; 1–4.

²⁴ Bar Nathan, *The Pottery*, 111.

²⁵ Because she accepts de Vaux's notion that there was a break in occupation which she dates to 9/8 BCE on the basis that that was the date of the most recent coin in a hoard found "buried beneath the level of Period II and above that of Period Ib" (de Vaux, *Archaeology* 34). She makes the curious statement that "de Vaux's description

with those from L 130 (which are actually earlier than those from L 89) she deduces that there is “a tendency for the walls of the deep cups or beakers with flaring rims to become thicker during the course of the Herodian period”²⁶ when, arguably, the reverse is true.²⁷

The Aqueduct from its sources northwest of the inhabited buildings

Two different building techniques are noticeable in the aqueduct which almost certainly represent temporal stages in its construction. The earliest stage is a simple, wide channel running down from below the cliffs to the site.²⁸ This aqueduct was a run-off aqueduct gathering rain water falling on the slopes northwest of Qumran.²⁹ At a later stage a dam was built across the foot of a waterfall in Nahal Qumran which, for a few days in some years, drains rain water falling in the Judean mountains. A technically more challenging aqueduct, which had to tunnel through the cliffs in some places, carried water from the dam to the beginning of the earlier aqueduct.³⁰

The first stage of the aqueduct, which relied entirely on limited local rainfall, probably already related to the “lower” channel (of “period Ia”), feeding only loci 110, 117 and 118. Following the earthquake, water gathered by the run-off aqueduct was channelled into the “main” channel which was built below floor level into the earthquake debris. Once the dam was built, and providing the Nahal flooded, a greater volume of water could be gathered and distributed into larger reservoirs.

The aqueduct in Loci 130, 132, 135 etc.

How this area gathered and fed water into the lower, ‘Phase Ia’ channel is unclear but there were two distinct stages in the aqueduct that

of the context makes it clear that the hoard could equally be associated with Period Ib, and common sense (*sic*) suggests this is the case. Hoards are often buried in times of trouble. . . .” (Magness, *Archaeology of Qumran*, 67) though she does not explain how one buries a hoard above the floor.

²⁶ Magness, *Debating Qumran*, 53

²⁷ R. Bar Nathan, *The Pottery* 98.

²⁸ Ilan and Amit, “Aqueducts,” 385; Fig. 1 no. 16 eastward.

²⁹ “A wall serving as a kind of low dam was built attached to the aqueduct, evidently in order to catch run-off” (Ilan and Amit, “Aqueducts,” 385). This section of the aqueduct still gathers run-off water, as it would have done before Nahal Qumran was dammed; see <http://www.bib-arch.org/BARExclusive/bswbBARMainPage.asp>.

³⁰ Ilan and Amit, “Aqueducts,” Fig. 1; cf. sections D-D and E-E.

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later crossed the area. In the first stage the decantation basin, L 132, was delimited by a wall to the east (to L 135) and to the south (to L 130). The main aqueduct channel started in the southeast corner with its floor raised slightly above the level of the basin, presumably to deter silt entering the channel. Neither the sides of the channel, nor the wall between L 130 and L 132, look as though they were built as free-standing walls³¹ (Pls 261–263). (Cf. Plate 1) Thus the floor associated with the aqueduct would be one covering the pottery and soil deposited in L 130, as a back-fill, “due to a clearing of the building after the earthquake” (L 130, 2/3/55). De Vaux interpreted a sediment found in L 130³² as being due to a dislocation of the aqueduct. He writes that, in the northwest, “it reaches a thickness of 75 cm. As it extends to the east it grows progressively thinner;” in his notes he quite rightly asked himself “why is the sediment not horizontal?” (L 130, 3/3/55). This, justifiably, continued to worry him (see L 130, 7/3/55) but he seems to have dismissed his qualms over time and convinced himself of his early hypothesis that “an earthquake broke down the water system.”

De Vaux clearly did not consider the extreme unlikelihood that any water would have been flowing in an aqueduct in Qumran at the time of the earthquake of 31 BCE. According to Josephus it occurred soon after the battle of Actium which took place on September 2 (*Ant.* 15.121). The sparse rain season during which the wadi might have running water is most likely to occur between February and April. However well constructed a dam, and however elaborate the system of sluice gates leading from it into the aqueduct, it is inconceivable that dammed water would have survived five or six months of the searing heat of summer in sufficient quantities to be discharging into an aqueduct in September or October.

I suggest that the sediment was not, in fact, from a recent flood but was rather a small part of the Lisan marl, the natural “bed-rock” in Qumran, which was dug out when the massive pool L 91 was created.³³ It was spread, in places, over the ashes from the fire that had resulted from the earthquake.³⁴ This was done as part of the first stage

³¹ The scrappy wall between L 130 and L 132, if freestanding, would not have been able to withstand the pressure of standing water.

³² De Vaux, *Archaeology*, 23

³³ The greatest thickness of the sediment against the wall separating L 130 from L 122 at the northwest and its tailing off to the east would be compatible with a dumping from the roof above L 122 accessed from the newly built staircase in L 113.

³⁴ De Vaux himself came to a similar conclusion when he encountered a “nearly sterile layer . . . upon a level of ashes” in L 66 (L 66, 2/3/54).

in the construction of the main aqueduct. Whether the wall that separated L 132 from L 135, which “was not so thick and seemingly without a foundation” (L 135, 6/3/55) could have been freestanding is also doubtful.

The large decantation basin L 132 could not have been very efficient and was probably soon replaced by a much smaller one in the north, L 136, and the start of the channel was extended northward to connect to the smaller basin. At the same time a stepped pool L 138 was built. This stretch of the channel was not freestanding and the area of L 132 must have been backfilled with a “fill of fine earth with few sherds” (L132, 7/3/55), which may have originated in L56/58 or L 71. An indication that this was so is given by the pool L 138, which supposedly received its water from the basin L 136.³⁵ The inlet to L 138 was at the level of the third step down into the pool.³⁶ The top of the steps clearly was the new “ground level” and this can be observed to have been the elevation of the top of the main channel where it exits L 136. (Cf. Plate 2) If the main channel had been freestanding there would have been no need to raise the sides of the pool superfluously above the inlet channel to include the two top steps as the pool could be filled no higher than the inlet.

Apart from the western buildings clustered around the iron age cistern, locus 110, there are few buildings that can definitely be assigned to a pre-aqueduct (and pre-earthquake) phase.

Pre-aqueduct activity to the south-east of locus 100

As already mentioned, when pools loci 91 and 56/58 were dug (of which at least locus 91 was dug contemporaneously with the construction of the aqueduct) some of the spoil was almost certainly spread around them thereby raising the neighbouring floor levels. A pre-aqueduct floor was encountered in locus 79: “In locus 79 we found a floor lower than the floor uncovered previously [which is presumably that associated with the ‘main’ channel]. [...] In this locus we distinguished four successive phases” (L 79, 14/3/54) of which at least one must surely have been pre-aqueduct.

³⁵ “What lies between the basin and L138 remains obscure” (L 137 9/3/55).

³⁶ K. Galor, “Plastered Pools. A New Perspective,” *Khirbet Qumran et ‘Ain Feshka. II Studies of Anthropology, Physics and Chemistry*. (eds. Humbert and Gunneweg; Göttingen: Academic Press Fribourg 2003), 291–320; see fig. 3.

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Pre-aqueduct activity in the south east of the so-called 'Main' building

The two kilns in L 66 clearly pre-dated the pools, loci 48–50, which were built over them and were fed by a branch off the main aqueduct. Accessible from the area of the kilns via a door were two pools, L 34, which seem to have existed before the aqueduct [“We excavated an earlier wall in front of the ‘threshold’ of L 36. . . . The wall founded on the ground was contemporary with the south wall of L 36 (and of L 34) and was earlier than the channel” (L 36, 13/3/56)]. The westernmost pool has some resemblance to free standing bathtubs in Jericho, particularly to the pair in the ‘Twin’ Palaces which had heating devices.³⁷ The stone lined aperture in the wall separating this room from the kilns and identified as a drain (poorly visible in Pl 73) was, perhaps, a flue from a furnace near the kilns, that heated water in a container standing in the area to the west of the flue where a pool is conjectured although no signs of plastering show up in any of the pictures. Could the stepped pool have gathered rainwater from a roof to the south and supplied water to the heating unit and tub? The pools do not have any identifiable means of obtaining water from the channel to the north of loci 56/58 [“it is not from L 67 that the basins of L 34 were fed unless it was made by an outlet into the basin at the southwest corner of L 34” (L 67, 27/2/54) for which no evidence exists. Moreover “the channel does not belong to the early stage, backed up as it is against a wall resting on the ground” which refers to the wall built partially on the remnants of the earlier wall against which the pools in L 34 were built (L 42, 12/3/53)].

Although the apparently pre-aqueduct wall was found to the south of loci 34 and 36 (shown on Humbert and Chambon’s Plan IV) there is no evidence that the excavators attempted to follow any westward continuation below L 30, nor are the implications of this early wall discussed. Further west “two soundings were made in order to verify that there was not a lower floor” (L 2, 28/3/55), and, as no mention is made of an early wall, it is probably safe to assume that it did not continue this far west. As there were no walls going to the north of the early wall it is logical to reconstruct returns to the south at the western and eastern extremities of what later became loci 56 and 58. The space between this “early” wall and the wall north of L 77 could have been divided into a pair of store-rooms.

³⁷ Netzer, *Palaces Vol II*, figs 220, 221, 225, 226.

Evidence for a pre-earthquake phase in the tower

In L 10A was found fallen mud-brick, possibly dislodged by an earthquake (Pl 18) and beneath it pottery all of which seems to pre-date 31 BCE.³⁸ All walls coming to the outside of the tower appear to abut it and could thus post-date it.

Evidence for pre-earthquake activity in the "Main" building

There is little evidence for post-Iron Age, pre-earthquake occupation in the so-called "Main" building. De Vaux accounted for this by maintaining that most of the rooms were completely cleared of any occupational and earthquake debris.³⁹ This is a most unlikely scenario. These buildings did not have impressive stone or mosaic floors to uncover, merely floors of beaten earth. It would have been far easier to create new floors covering the debris as was done in the "Period Ia" building. Moreover the paucity of the water resources makes it unlikely that much of the "Main" building would have been constructed before the aqueduct was built.

L 2. Some of the pottery forms found in L 2 first appeared before 31 BCE. This is particularly true for the "Hasmonean" cooking pot⁴⁰ but these continue to be found, albeit in smaller numbers, until the end of Herod's reign.⁴¹ The latest coins found on the earliest floor, however, are dated to Herod Agrippa (41–44 CE). Moreover the south-east corner of L 2 is shown on plans as though it was built over, and thus contemporary with, or later than, the channel skirting loci 56/58.⁴² In the field the visible corner stones of L 2 can be seen to project over the path of the channel but how the channel relates to the foundations of the walls can not be seen.

The Sequence of aqueducts and pools east of L 91

Whilst stages can be detected in the construction of the aqueduct and pools to the east of L 91 (actually of its settling tank L 83) it is not possible to say if the earliest stage was part of the original "main

³⁸ De Vaux, *RB* 61: Figs 1:2, 3:2, 5, 6, 10, 12, 15, 21, 22)

³⁹ De Vaux *Archaeology*, 25; Magness, *Archaeology of Qumran*, 68.

⁴⁰ R. de Vaux, "Fouilles au Khirbet Qumran: Rapport préliminaire," *RB* 60 (1953), Fig 3:7.

⁴¹ Bar-Nathan, *The Pottery*, 69.

⁴² In particular see de Vaux, *RB* 61, pl V.

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aqueduct” or a later addition although it is unlikely that there would have been a long delay in providing maximum storage capacity for the precious water that the new aqueduct delivered to the site. The massive pool L 91 was certainly dug as part of the main aqueduct and may have been its original termination. As rainfall in Qumran is unreliable it would have been difficult to judge the potential volume of water that could be garnered. If, in the following winters, L 91 was being filled to overflowing, then further reservoirs would quickly have been added. There were at least two stages in the construction of the aqueduct that branched off from L 83. In L 54 “We removed the upper floor and uncovered a conduit which branches from the west-east channel. . . .” “It belongs to Level I. At level II (or before . . .) it was abandoned. Its point of departure from the main channel is closed off (it seems also that the main channel had then been raised)” (L 54, 25/3/54).

The west-east channel penetrated the wall to the south of L 54. This wall (which for simplicity I will call “W1”) was considered to be a retaining wall of iron age origin⁴³ and belongs to the pre-aqueduct phase as it is an eastward continuation of the southern boundary of L 102. The main channel continued eastward to feed pools L56/58 and, via an early channel which by-passed L56/58, to pool L48/49⁴⁴ [In L 42, “In front of L 30, under the upper channel, which begins at the exterior corner of L 2, is a lower channel. Therefore there are two stages to the channel, which itself (cf. supra) must already be of second stage” (L 42, 13/4/53)].

When the kilns in L 66, which would have stood in the open, were in operation the wall that separated them from loci 34, 37, etc., would have been the eastern extremity of the built up area. The pool L 48/49 (and loci 51, 39, etc.) were built over the kilns and at a higher level than the threshold of the door into L 34 (see above) after a parallel wall further to the east (that between L 51 and L 44) had been built and backfilling (“two floors were separated by more than a meter of fill” L 39, 9/4/53) had taken place utilising spoil from the digging of L 56/58.

As the aqueduct was not built before the earthquake of 31 BCE the pool L48/49 could not have existed before that date. Thus the cracks visible in the pool (Pl 163) could not have been caused by the quake

⁴³ De Vaux *Archaeology*, 2.

⁴⁴ As noted above the channel was built against a later wall than that south of pools loci 34 and 36 so it was not built to take water to them.

of 31 BCE as claimed by de Vaux.⁴⁵ If they were indeed caused by an earthquake, rather than by the settling of insufficiently consolidated backfill surrounding the pools, then it is more likely they were the result of the earthquake of 48 CE⁴⁶ which would help account for the coin of Herod Agrippa (37–44 CE) which was found close to the bottom of the cistern (L 49, 21/4/53).

At the pre-aqueduct stage the room, L 77, south of W1 did not exist. Observation in the field (and cf. Pl 319) reveal that the western wall of L 77 is not bonded to W1 (Cf. Plate 3) and is thus a later addition.⁴⁷ Before L 77 was built the ground level south of W1 must have sloped down from west to east. The elevation at the west would have been higher than the floor level of L 77, close to that extant in loci 81 and 86, whilst at the eastern end the level would have been lower, close to that in L 73 (Pl 182). The ground in front of the door in W1 was probably flush with the base of the lowest step that was integrated into the doorway (Pl 324, base of the third step up). The channel from L 54 running through wall W1 was inserted before room L 77 was added and was at the then ground level (Pl 327). It may merely have drained excess water into gardens south of the built-up area but there is a decided possibility that it carried water in a channel south of wall W1 to a row of three small pools, loci 72, 68 and 70. (Cf. Fig. 2) Although a channel which fed L 68 after running north of L 72 is shown on the plan⁴⁸ as branching off the stretch of aqueduct between the north east of L 58 and the south east of L 77, no mention is made of this junction in the field notes and it is not visible in the limited photographic record (Pls 365, 366); nor is it visible any longer in the field. (Cf. Plate 4) Moreover one photograph (Pl 365) seems to indicate that this channel into L 68 was at a lower level than the “main” channel and thus may have run beneath it. It would have been illogical for such a small pool as L 68 to have had two feeder channels from the same water source (as shown on Plan IV), thus the channel into its north west corner probably pre-dates the “main” channel and connected to the channel in L 54 that broached W1.

We have already seen that the channel that by-passed L 56/58 to the north had two phases. No indications were given that the channel running from the north east corner of L 56/58 had more than one

⁴⁵ De Vaux *Archaeology*, 27; and see Magness, *Archaeology of Qumran*, 64.

⁴⁶ D. Kallner-Amiran, “A Revised Earthquake Catalogue of Palestine,” *IEJ* 1 (1950): 225.

⁴⁷ This had been brought to my attention previously by Ian Hutchesson.

⁴⁸ Humbert and Chambon, *Fouilles*, Plan IV.

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phase. It thus must relate only to the second, raised level of the bypass, which in the earlier, lower phase terminated at L 48/49. De Vaux believed that this section of the aqueduct, west of loci 66 and 73, rested on a fill from the digging of cistern L 71 (L 66 2/3/54 and 9/3/54; and see <http://www1.ku-eichstaett.de/KTF/qumran/bilder/abfrage/loc66.jpg>) which must have post-dated loci 72, 68 and 70.⁴⁹ This extension of the channel was built at the same time as L 77. A break had to be made in W1 east of L 77 to allow the channel to pass through (L66, 3/3/54) and the partial bonding visible today was possible between the eastern wall of L 77 and the rebuild of the break in W1.

When room, L 77, was built some of the ground at the western end had to be dug out and dumped at the eastern end to create a level floor surface. This necessitated the removal of the projected continuation of the channel from L 54⁵⁰ and the addition of the two extra steps in front of the doorway in W1. The supposed slope of the floor⁵¹ was not intentional but was the result of the loose soil in the east settling over time.⁵² Although de Vaux interpreted locus 77 as a communal refectory it should be compared, in size, shape, and function with storerooms in Jericho,⁵³ Herodium,⁵⁴ and Masada. It would have replaced the storerooms conjectured above to have existed in the area later occupied by pool L 56/58 before the construction of the “main” aqueduct.

The small storeroom, L 86/89, seems to have been a later addition to the south west end of L 77. Although the point at which the walls met with the southern wall of L 77 was disturbed by the introduction of the Period III channel (L 86, 16/3/54) they do not appear to have

⁴⁹ Although Galor suggests that loci 72 and 70 went out of use after L68 was constructed (Galor, *Plastered Pools*, 301), I believe that they were abandoned after their original water source was replaced by the “main” channel to which only L 68 was connected.

⁵⁰ Remains of a channel might survive buried beneath the eastern fill.

⁵¹ De Vaux *Archaeology*, 11.

⁵² It is possible that the roof of L 77 was originally supported by a line of wooden posts running at intervals down the centre of the room. As the floor settled at the eastern end the mud-brick pilasters had to be inserted either to completely replace the posts or to support their bases. The channel, even if it co-existed with the floor of L 77, would only have carried water for a few days in the rainy season, so it could scarcely have been used “to clean the room at frequent intervals” (de Vaux, *Archaeology*, 11). Moreover, unless there was an effective sluice gate, the water would have run uncontrollably and at some speed and would rapidly have eroded any ordinary plaster floor.

⁵³ Netzer, *Palaces* 1.131–132, plan 14.

⁵⁴ E. Netzer, *Greater Herodium* (Qedem 13; Jerusalem: Hebrew University 1981) 22–25.

been bonded originally (Pl 348). The pottery vessels found in locus 89 must thus belong to a period at least two sub-phases later than the construction of the main aqueduct in 31 BCE. Although they were claimed to be from the pantry of a community dining room, they were more likely to have been a store of locally made vessels awaiting sale, as was suggested by Donceel.⁵⁵

Qumran in the Hasmonean period

Apart from the tower which was probably built (on Iron Age foundations?) to watch over the path up to Hyrcania, Hasmonean Qumran consisted almost entirely of pottery kilns, workshops or storage areas (loci 66, 101, 120, 121, 122, 125). There was little surviving evidence for domestic activity beyond the “bathtub” in L 34, an oven that predated L 30 (L 30, 28 & 30/3/53) and, possibly, a small oven in L 100 (29/3/55). The water supply was meagre and it is probable that the occupation of the site was confined mainly to the winter season when the water was at its freshest and most abundant. There was insufficient water for agriculture and the storerooms would have been necessary to hold sufficient dry goods to supply the seasonal potters, lye makers,⁵⁶ salt and bitumen gatherers, and other artisans who were probably based at the large agricultural and processing royal estate in nearby Jericho.

The Jericho estate was founded by John Hyrcanus (134–104 BCE) or possibly his father Simeon (143–134 BCE)⁵⁷ but would have begun to flourish under Alexander Jannaeus (103–76). He, who was “during his reign of 27 years, almost continuously involved in foreign and internal wars, for the most part deliberately provoked by him,”⁵⁸ would early have recognised Qumran’s strategic location, militarily and commercially; not only did it control the path up to Hyrcania, but it overlooked both the track to Ein Gedi and the northern end of the Dead Sea and any boat traffic plying upon it. It is possible that Jannaeus intended to build a substantial desert fort in Qumran whose

⁵⁵ R. Donceel, “Reprise des travaux de publication des fouilles au Khirbet Qumran,” *RB* 99 (1992) 557–73.

⁵⁶ Z. Amar, “The Ash and the Red Material from Qumran,” *DSD* 5 (1998) 1–15. It should be noted that soda ash, as well as being used in the manufacture of soap, is also used in cloth dying and glass manufacture.

⁵⁷ Netzer, *Palaces*, 1.1.

⁵⁸ E. Schürer, *The History of the Jewish People in the Age of Jesus Christ* (eds. G. Vermes, F. Millar, and M. Black; Edinburgh: T&T Clark, 1973) 1.220.

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inhabitants could also have exploited the area for irrigated agriculture. All the Hasmonean desert forts, however, had substantial rock-hewn, underground water cisterns in which water could be stored in the best conditions to minimise evaporation. As no such cistern was constructed in Qumran the indications are that any such plans were interrupted, probably by the same circumstances—the threat of invasion from the Nabateans around 94 BCE (*Ant.* 13.375; 13.392) that, in Jericho, led to the hurried construction of the fortified palace.⁵⁹ With the expansion of Jannaeus' sphere of control east of the Jordan, and particularly following the construction of Machaerus, c. 90 BCE (*War.* 7.6.2) the strategic importance of Qumran was considerably reduced.

The early Hasmoneans were, however, in need of a steady and reliable income—John Hyrcanus had had to plunder King David's tomb to pay his mercenaries (*Ant.* 13.249)—and the plains west of Jericho, once irrigated with water brought by aqueduct from Wadi Qelt, supplied them with one. As their agricultural endeavours flourished, the Hasmoneans, if they were to expand their productivity and increase their income, had to look for ways to bring more land into production. Before the water in the Qelt aqueduct had been fully exploited (late in the reign of John Hyrcanus or early in that of Alexander Jannaeus) they would have begun to scour the neighbourhood either for alternative water sources that could be brought to Jericho, or for nearby locations suitable for growing dates or balsam. Even if none could be found they would have done their best to minimise the use of the “expensive” water supply in Jericho for other than agricultural and domestic purposes.

Eventually, to increase productivity in Jericho, Jannaeus built the technically demanding aqueduct that brought spring water from Na'aran, supplemented by a channel from Ein el-'Auja, some 13 km to the north of Jericho;⁶⁰ that is the same distance to the north of Jericho as Qumran is to the south. Qumran must be considered an integral, though outlying, part of the Royal estate.⁶¹

⁵⁹ Netzer, *Palaces*, 1.3–4.

⁶⁰ E. Netzer and G. Garbrecht, “Water Channels and a royal estate of the late Hellenistic period in Jericho's western plains,” *Aqueducts of Israel*, 367–379, esp. 372; Netzer, *Palaces* 2.6–8, Plan 2.

⁶¹ Bar-Adon came to a broadly similar conclusion when the excavation of Second Temple Jericho was in its infancy; see P. Bar-Adon, “The Hasmonean Fortresses and the status of Khirbet Qumran,” *Eretz Israel* 15 (1981) 349–352 (Hebrew), 86* (English summary). More recently Drori and Magen concluded that “the founding of Qumran should be viewed as an integral part of the Hasmonean plan to settle and

Establishing and running the expanding estate could only have been achieved with a large influx of labourers and artisans who would all have required a supply of mundane domestic pottery, the manufacture of which in Jericho itself would have used both precious water and fuel—another commodity in short supply in that barren land. Kilns were therefore established in Qumran, utilising run-off water collected in the renovated Iron Age cistern L 100.⁶² De Vaux firmly dated two pottery kilns (in L 66) to the earliest post-Iron Age occupation, adding “there is nothing to indicate that these kilns were already in service during the Israelite period.”⁶³ From the nearby Dead Sea bitumen which, mixed with dried reeds from the extensive reed beds at Ein Feshka, palm fronds, any available wood, and the usual fuel of the desert, dung, could have been used to fire them. Although they would have produced noxious gases, locating the kilns in Qumran would have ensured the fumes were well away from the royal palace.

The storehouses were part of the supply chain for distributing food, originating particularly in the Galilee, to Ein Gedi, and to the fortresses Machaerus and Hyrcania, before the latter were destroyed by the Romans in c. 56 BCE (*Ant.* 14.89). They would have held sufficient dry goods for the seasonal artisans coming in from Jericho and for caravans passing between Ein Gedi and Jericho. Whilst some initial processing of the balsam plant may have taken place in Ein Gedi the final manufacturing processes whereby valuable unguents, perfumes and medicines were produced, were secrets probably retained, as far as possible, by the royal estate in Jericho. Qumran was a convenient caravanserai at which partially processed material could break its journey northwards to Jericho for final production and sale.

Qumran’s position as an entrepôt depot would have increased in importance when Herod not only rebuilt Hyrcania and Machaerus but developed Masada as a fortress and luxurious desert retreat. Soon after the Battle of Actium early in September 31 BCE Herod, who had

fortify the Jordan Valley” (M. Wise, M. Abegg and E. Cook *The Dead Sea Scrolls: A New Translation* [San Francisco, Harper 1996], 22–23). This concept is echoed by Hirschfeld (Hirschfeld, *Qumran*, 288). Cansdale has indicated “a close connection between Jericho and Qumran with a distinct possibility that the Jericho priests controlled the settlement” (L. Cansdale, *Qumran and the Essenes. A Re-Evaluation of the Evidence* [Tubingen. Mohr, 1997] 190). For more on the agricultural importance of the Jericho estate see D. Stacey, “Hedonists or pragmatic agriculturalists? Reassessing Hasmonean Jericho,” *Levant* 38 (2006) 191–202.

⁶² De Vaux, *Archaeology*, 4.

⁶³ De Vaux, *Archaeology*, 4–5

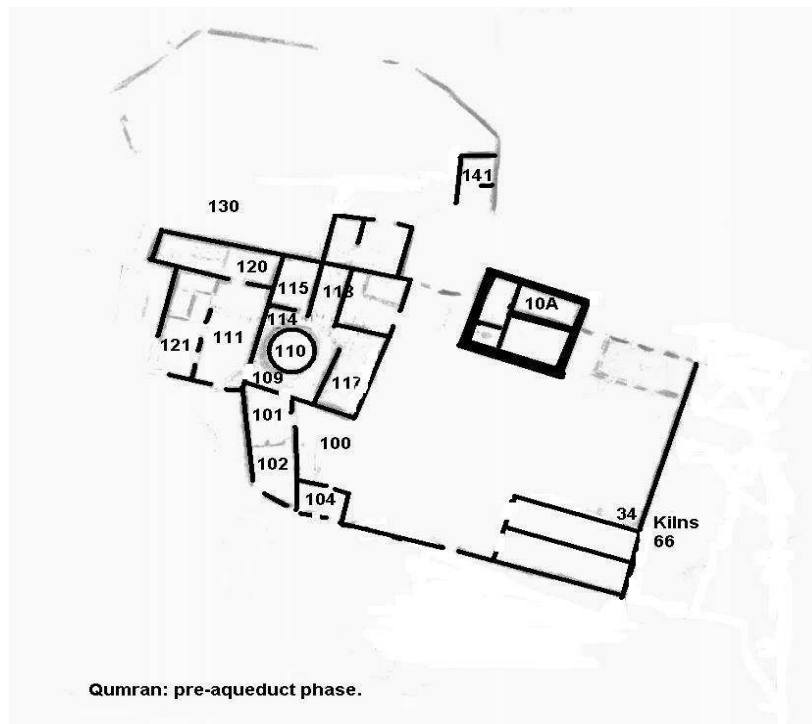
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supported Anthony and the losing side, decided that he had to visit the victor, Octavian, in an attempt to salvage his position. Not being confident of his complicated family loyalties whilst he was away he banished his wife, Mariamne, and her mother to Alexandrium, but sent his two young sons, under the watchful eye of his mother, to Masada (*Ant.* 15.183–86), of which little, at that time, had been built. Qumran was well located to facilitate, amongst other things, the collection and transportation of building materials, and food supplies suitable for his royal sons, to Masada. However, the earthquake of 31 BCE, which occurred soon after the battle of Actium (*Ant.* 15.121), would have disrupted the water supply of Qumran and necessitated the rebuilding of the aqueduct system. Moreover it needed to be expanded so that sufficient water could be stored to make some year-round occupation possible. Although the run-off aqueduct would still have gathered water there was an immediate need to build the aqueduct over the earthquake debris in L 114–15, and to clear out and increase the capacities of loci 100, 117 and 118, before the start of the sparse rainy season which is most likely to occur between February and April. The first stage of this aqueduct probably terminated in the large pool L 91. The building of the dam across Nahal Qumran and the tunnelling of the aqueduct from it through the rock-face could have commenced at the same time but is unlikely to have been completed in time for that winter's rains. It is probable that the pools to the east of L 91 would not have been added until it was functioning in the next winter or two.⁶⁴

⁶⁴ This article was written and submitted before I had seen K. Galor, J.-B. Humbert, and J. Zangenberg (eds), *Qumran: The Site of the Dead Sea Scrolls: Archaeological Interpretations and Debates* (Leiden: Brill, 2006).

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Plate 1. The channel looking north from L 130/131. The two side walls of the channel, particularly the western one, show signs of recent restoration/conservation. However the large flat stones placed vertically in both walls seem original and were surely not built into very narrow freestanding walls but into retaining walls in the side of a channel below floor level.

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Plate 2. Looking north from L 131 towards L 138 beyond the modern walkway.

The floor level associated with L 138 must have been above the visible outlet (beyond the walkway) and would have reached to the top of the aqueduct channel at much the same elevation as the modern walkway.

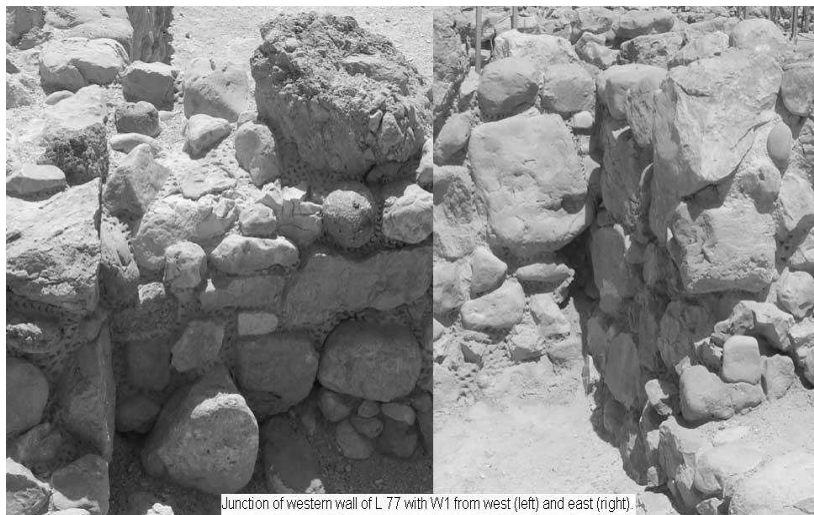


Plate 3. Junction of western wall of L 77 with 'W1' from the west (left) and the east (right).

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Plate 4. Looking west from L 68 (lower left)
towards L 77 (notice board).

No signs of the early channel into the northwest corner of L 68, still visible in Pl 365 survive today. However the stones visible behind the fence post below the surviving channel may be from the proposed earlier channel.